



THE UNIVERSITY OF BRITISH COLUMBIA
Faculty of Forestry

Dear Dr. Berenbaum,

Please consider our revised manuscript, “Solstice optimizes thermal growing season” (#2025-06796) for publication as a Brief Report in *PNAS*.

In this work, we explore the recent hypothesis that the summer solstice may be a universal cue for major plant physiological transitions^{1,2}. Using a thermal trade-off framework, we test this idea across Europe and North America, and over past, present, and future climates. Our results show that while the solstice often aligns with a thermal optimum, there are substantial regional variations that suggest more locally adaptive cues. This work brings new perspective to the solstice debate and highlights promising directions for future research, with key implications for understanding the drivers of phenological transitions and forecasting plant responses to climate change.

Both reviewers and the editor recognized the significance of our findings. Their comments have greatly helped us improve the manuscript and have strengthened the overall contribution of our study, across research fields from physiology, ecology, and climatology.

In revising the manuscript, we were careful to more clearly acknowledging the limitations of our approach—especially regarding other climatic constraints, such as moisture availability, that influence the growing season. We also clarified the scope of our study, specifying that our analyses focus on temperate regions in the Northern Hemisphere. Following this, we modified the two main figures by adding inset panels to show that our findings are not limited to Europe, but also hold across North America.

We detail our changes, point-by-point, in the following pages. Both authors have contributed to and approved of this submission, and the manuscript is not currently submitted to or in review in an other journal. We hope that you will find it suitable for publication in *PNAS*.

Sincerely,

A handwritten signature in blue ink that reads 'V. Van der Meersch'.

Victor Van der Meersch, PhD
Forest & Conservation Sciences
University of British Columbia

References

- [1] C. M. Zohner et al. *Science* (2023).
- [2] V. Journé et al. *Nature Plants* (2024).